

Pongporn (Ken) Supa

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Education

University of Bristol

Bristol, UK

MSc in Robotics (74/100, Expected Distinction)

Sept 2024 – Sept 2025

- **Thesis:** Whisker-Inspired Tactile Sensing for Bumper Drone: designing elastic morphology for aerial physical interaction under supervision of [Dr. Basaran Bahadir Kocer](#) [🔗](#)
- **Selected Coursework:** Swarm Robotics, Robot Learning and Teleoperation, Soft Robotics, Machine Vision, Robotics Fundamentals
- **Research Interests:** Autonomous navigation, Motion planning and Control, UAV design

Chulalongkorn University

Bangkok, TH

BEng in Mechanical Engineering (3.63/4.00, First Class Honour)

Aug 2017 – Aug 2021

- **Selected Coursework:** Mechatronics (A), Automatic Control Systems (A), Dynamic System Simulation (A), Statistics (A), Linear Algebra and Differential Equation (A), Computer Programming (A)

Work Experience

Robotics Software Engineer

Bangkok, TH

AI and Robotics Ventures, HORRUS project

Apr 2022 – Aug 2024

- Developed autonomous UAV system achieving 95% reliability in continuous surveillance operations by engineering a robust and real-time middleware using MAVLink protocol with asynchronous programming
- Designed and implemented a coverage path-planning algorithm for aerial photogrammetry with constraint-based geofencing
- Integrated real-time video streaming and object detection pipelines using Python, ROS2, and OpenCV for security and inspection drones, securing £7K in government funding
- Built UAV back-end systems with embedded communication protocols and API interfaces for multi-agent coordination

Research Assistant: Modular Temperature-Controlled Syringe Pump Extruder for Soft Material 3D Printer under [Assoc. Prof. Ratchatin Chancharoen](#) [🔗](#)

Bangkok, TH

May 2021 – Oct 2021

Department of Mechanical Engineering, Chulalongkorn University

- Transformed a cartesian robot into the syringe-based soft material 3D printer, which is capable of controlling the temperature of the syringe and printing bed, helping secure £67K funding from the National Research Council of Thailand (NRCT)
- Designed PCBs for soft material printing benefits, including a high-temperature extruder, dual-temperature controller printing bed, and nozzle calibration module to address complex behavior of viscoelastic materials.
- Extended open-source 3D printer firmware to create custom control interfaces and generate feature-based trajectories for adaptive printing based on real-time shape detection, leading to the prototype showcase at Thailand Inventor's Day

Selected Projects

Adaptive Point-Tracking Control of Multi-Robots Box Pushing

[Github repo](#) [🔗](#)

- Developed adaptive control algorithms for cooperative heterogenous mobile robots
- Implemented PID controllers and sensor fusion for accurate mobile robot coordination

Learning from Demonstration on Robot Manipulation

Mar 2025 - Apr 2025

- Applied Dynamic Movement Primitives (DMPs) and Gaussian Mixture Regression (GMR) for trajectory imitation learning
- Designed PD controller for Leader-Follower Teleoperation system

Human-like vs. robot voice. Who gives better encouragement?

Mar 2025 - Apr 2025

- Integrated Text-to-Speech AI and Co-verbal gestures to NAO robot, enabling human-like behavior

Search & Rescue in Mobile Robot

[Github repo](#) [Youtube](#)

- Designed and Developed Finite-State Machine to perform sensor calibration routine, obstacle avoidance and search and rescue operation on a known map
- Implemented dead reckoning method for localisation, and pose correction using prior knowledge of the environment
- Employed PID speed controller on mobile robots and low-pass filter in speed estimation

Apple Counting in Orchards

Dec 2024 - Jan 2025

- Designed and implemented image processing pipeline and YOLO model using color, geometric, and texture features under varying illumination using OpenCV and PyTorch

Teachings

Teaching Support in Visual Analytics Course

Jan 2025 - May 2025

- Demonstrated and led tutorial sessions on data visualisation and statistical methods

Teacher Assistant in Manufacturing Process Course

Mar 2022

- Prepared courseware/engineering workshops. Lectured on how 3D printing settings and parameters affect printing quality

Skills

Programming languages: Python, C/C++, MATLAB, Simulink, Javascript

Software: ROS2, OpenCV, PyTorch, PX4, QGroundControl, RViz, Gazebo, Docker, AWS EC2/ECR/S3

Hardware: Proficient in digital fabrication for rapid prototyping, UAV prototyping, PCB design, CAD/Fusion360, embedded systems

Honors & Awards

Bristol Award Plus, *University of Bristol*

Mar 2025

Think Big Scholarship Award Recipient, *University of Bristol*

Sept 2024

PTTEP Innovation Award, Silver Award, *Autonomous Offshore Drone System*

July 2024

Spotlight on New Technology Awards, *OTC Asia 2024, AI and Robotics Solutions for Offshore Digitalisation*

Feb 2024

Invited speaker on Robotics & AI Bootcamp, *Chula Engineering NITAD 18*

Mar 2022

References

Dr Basaran Bahadir Kocer

Lecturer

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